

Section II — Case III:
Defined Contribution Plan and MPF Retirement
Planning

AMA1D03 Introduction to Pension Mathematics

June 15, 2026

(a) Descriptions of the Defined Contribution Plan and Its Application in the Pension System

1. Defined Contribution (DC) versus Defined Benefit (DB)

Under a **defined contribution (DC)** plan, the **contribution side is fixed but the benefit is uncertain**. The employer and/or employee contribute a specified amount into an individual account; the terminal value depends on contributions, investment returns, and fees. Investment and longevity risks are borne primarily by the member.

By contrast, a **defined benefit (DB)** plan has a **fixed benefit formula but uncertain required contributions**. The sponsor uses actuarial assumptions to set funding levels and bears most investment and longevity risk if the plan is underfunded. Hong Kong's Occupational Retirement Schemes (ORSO) may be structured as either DC or DB, operating alongside mandatory MPF as part of the second pillar.

Hong Kong's Mandatory Provident Fund (MPF) System, launched on **1 December 2000**, is a compulsory DC scheme. It forms the **second pillar** of Hong Kong's retirement protection framework:

- **First pillar:** Flat-rate, means-tested cash benefits—CSSA (last-resort safety net), Old Age Living Allowance (OALA, for lower-income seniors), and Old Age Allowance (OAA, broader but smaller)
- **Second pillar:** MPF mandatory and voluntary contributions (plus ORSO where applicable)
- **Third pillar:** Personal savings, insurance, and family support

Hong Kong has **no earnings-related public pension**. First-pillar benefits are fixed amounts, so MPF is the core retirement base for the middle class—a structural difference from Canada (CPP) or the US (Social Security).

2. Types of MPF Schemes

According to the Mandatory Provident Fund Schemes Authority (MPFA), MPF schemes are classified into three main types:

Table 1: Types of MPF schemes

Scheme Type		Description
Master Schemes	Trust	Open to employees of participating employers and self-employed persons
Employer-sponsored Schemes		Only for employees of a single employer
Industry Schemes		For employees in specific industries (e.g., catering, construction)

Each employee has a personal account within a scheme. When changing jobs, accrued benefits are portable and can be transferred to the new employer's scheme or retained in a personal account.

3. Mandatory and Voluntary Contributions

Mandatory Contributions (MC):

- Employee and employer each contribute **5%** of relevant income
- Relevant income is subject to minimum (HK\$7,100) and maximum (HK\$30,000) monthly thresholds
- If monthly income is below HK\$7,100, the employee makes no mandatory contribution, but the employer still contributes 5% of HK\$7,100
- Maximum mandatory contribution per party: HK\$1,500 per month

Voluntary Contributions:

- **In-scheme voluntary contributions (VC):** extra contributions within an MPF scheme; no salaries-tax deduction; withdrawal rules follow mandatory contributions
- **Tax Deductible Voluntary Contributions (TVC):** separate personal accounts; up to **HK\$60,000 per year** deductible for salaries tax; more flexible withdrawal conditions

In Case III, Peter and Mary each contribute HK\$1,000 (5% of HK\$20,000 salary), matched equally by their employers, giving a total monthly contribution of **HK\$2,000**.

4. Benefits, Withdrawal, and Retirement Age

Members can generally withdraw accrued benefits when reaching age **65**. Other withdrawal conditions include permanent departure from Hong Kong, total incapacity, terminal illness, early retirement at age 60 with cessation of employment, **small-balance withdrawal** (total accrued benefits not exceeding HK\$5,000), and **death benefits** payable to nominated beneficiaries.

Upon withdrawal, members may take a lump sum or convert part of the balance into an annuity through the MPF platform. Annuities are the primary tool for hedging **longevity risk**—the key vulnerability of DC plans where members outlive their account balances.

5. Investment Options, Returns, and Fees

MPF offers multiple fund types: Hong Kong, China, Asia, and global equity funds; mixed assets; bond and guaranteed funds; and index-tracking funds such as the Tracker Fund of Hong Kong.

Members bear **investment risk**. Fund managers charge management fees, typically 0.5%–2% per annum, which reduce net returns over long horizons. In Case III, Peter invests in a Global Fund while Mary invests in a Hong Kong Fund, illustrating how asset allocation affects long-term outcomes under the same contribution structure.

6. Tax Treatment and System Role

Mandatory MPF contributions are tax-exempt up to statutory limits. TVC provides an additional tax incentive for members seeking to close retirement income gaps. Academic and industry studies consistently show that mandatory contributions alone are **insufficient** to maintain pre-retirement living standards, so members are expected to supplement MPF with voluntary savings and, where eligible, government cash benefits.

(b) Compare and Analyze the Defined Contribution Plan — Peter and Mary Examples

1. Modelling Setup

We simulate monthly MPF accumulation from December 2000 until each member reaches age 65 using a Python model (`case3/main.py`). Key assumptions are summarised in Table 2.

Assumption notes and limitations:

- **Salary growth = CPI** implies **zero real wage growth**—a strong simplification. Real career wage growth would raise the final-salary denominator and lower the replacement ratio further.
- **Returns:** historical monthly returns are used through the latest available data; thereafter a **4% p.a. nominal** forward return is applied (consistent with long-run MPFA/industry planning assumptions). This is not a full historical back-test—outperformance of global funds over Hong Kong equities in 2000–2025 reflects a specific period and involves **hindsight bias**.
- **4% withdrawal rule:** retirement income equals 4% of the balance at retirement in the first year, with the implicit Bengen-style assumption that subsequent withdrawals rise with inflation. The rule was calibrated to US 30-year retirements; Hong Kong life expectancy exceeds 85, so a fixed withdrawal rate carries **longevity depletion risk**.
- **Fees:** 1% p.a. is mid-range for MPF (index funds can be below 0.5%; active equity funds often 1.2%–1.8%). Fee drag compounds over 25–45 years.

Table 2: Key modelling assumptions

Parameter	Value	Note
Start date	1 Dec 2000	MPF launch
Retirement age	65	Normal retirement
Initial monthly salary	HK\$20,000	Case III
Monthly MC (employee + employer)	HK\$2,000	5% + 5%
Salary growth	HK Composite CPI	Zero real wage growth
Hong Kong Fund proxy	Hang Seng Index (^HSI)	Tracker Fund replicates HSI
Global Fund proxy	Spliced ^GSPC/EFA/ACWI	Full history from Dec 2000
Forward return	4% p.a. (nominal)	After latest available data
Fund expense ratio	1% p.a.	Mid-range MPF assumption
Retirement income	4% withdrawal rule	Year-1 income = 4% of balance

Table 3: Case III participant settings

	Peter	Mary
Start age (Dec 2000)	20	40
Retirement age	65	65
Contribution period	45 years	25 years
Retirement year	2045	2025
Assigned fund	Global (spliced)	Hong Kong (^HSI)

2. Baseline Results

Table 4: Baseline simulation results

Indicator	Peter (Global)	Mary (Hong Kong)
Contribution years	45.1	25.1
Total contributions (HK\$)	1,443,931	723,931
Investment gains (HK\$)	3,552,431	126,930
Final MPF balance (HK\$)	4,996,362	850,862
Final annual salary (HK\$)	555,156	374,107
Annual retirement income (HK\$)	199,854	34,034
Monthly retirement income (HK\$)	16,655	2,836
Replacement ratio	36.0%	9.1%

Figure 1 shows Peter's balance rising steeply over 45 years, while Mary's balance grows more slowly over 25 years, ending at roughly one-sixth of Peter's balance. For Peter, investment gains (HK\$3.55 million) far exceed total contributions (HK\$1.44 million), demonstrating the power of long-horizon compounding in a DC system.

Historical Hong Kong equity returns (via the Hang Seng Index) were volatile over Mary's 2000–2025 contribution period, including the post-2000 downturn and 2008 financial crisis, which helps explain her lower final balance relative to earlier proxy estimates.

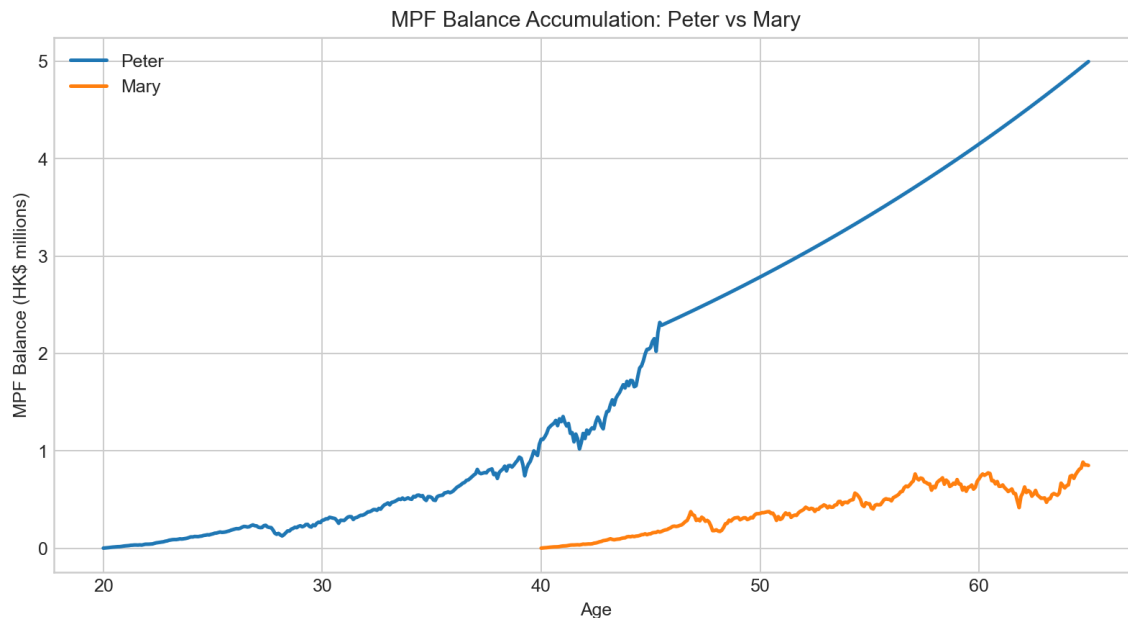


Figure 1: MPF balance accumulation: Peter versus Mary

3. Advantages and Disadvantages

Advantages:

1. **Clear property rights and portability:** Individual accounts suit Hong Kong's mobile labour market; no employer-credit risk as in DB plans.
2. **Employer matching:** The employer's 5% contribution doubles savings without additional employee cost.
3. **Investment choice:** Members can select funds aligned with risk tolerance.
4. **Tax incentives:** TVC of up to HK\$60,000 per year is tax-deductible.
5. **Compounding:** Early starters benefit enormously from decades of compound growth (the "Rule of 72": at 4% p.a., wealth doubles roughly every 18 years; 45 years yields $\sim 6\times$ compounding versus $\sim 2.7\times$ over 25 years).

Disadvantages:

1. **Investment and sequence risk:** Members bear full downside risk; a pre-retirement market crash can sharply erode terminal value.
2. **No risk pooling:** Unlike DB plans, there is no cross-subsidy for longevity or investment shortfalls.

3. **Low replacement ratio:** Even Peter achieves only 36% replacement from mandatory contributions, below the 60–70% commonly recommended.
4. **Fee drag and behavioural erosion:** Management fees compound negatively; poor fund switching further reduces net returns.
5. **Late starter penalty:** Mary’s 9.1% replacement ratio shows that starting at 40 cannot be fully offset by fund selection.

4. Start Age versus Fund Choice

Table 5 reports replacement ratios by start age. Table 6 reports fund-swap sensitivity at fixed start ages.

Table 5: Replacement ratio by start age and fund

Start Age	Fund	Final Balance (HK\$)	Replacement Ratio
20	Global	4,996,362	36.0%
20	Hong Kong	2,427,886	17.5%
30	Global	3,417,078	30.0%
30	Hong Kong	1,500,213	13.2%
40	Global	2,063,587	22.1%
40	Hong Kong	850,862	9.1%
50	Global	520,922	6.9%
50	Hong Kong	457,617	6.0%

Table 6: Fund-swap sensitivity

Name	Assigned RR	Swapped RR	Change (pp)
Peter (age 20)	36.0%	17.5%	−18.5
Mary (age 40)	9.1%	22.1%	+13.0

Key finding: Comparing start ages at the same fund shows a **27 percentage-point gap** (36% versus 9.1% for global and Hong Kong funds at ages 20 and 40). Peter at age 20 with the underperforming Hong Kong fund (17.5%) still ends with a far higher balance than Mary. **Contribution horizon dominates fund selection** because terminal value is exponentially sensitive to time and only power-sensitive to return.

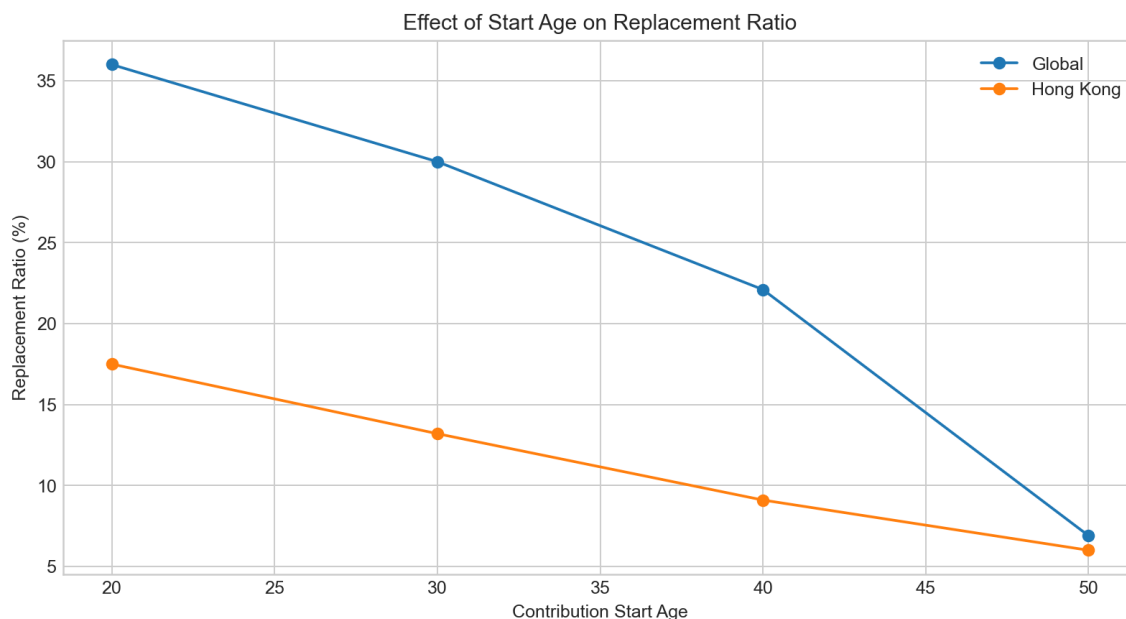


Figure 2: Effect of start age on replacement ratio

5. Living Expenses, Inflation, and Extra Contributions

Financial planners often target a **60–70% replacement ratio**. Using final monthly salary from Table 4 ($60\% \times \text{final monthly salary} - \text{MPF monthly income}$): Peter's gap is approximately HK\$11,103; Mary's gap is approximately HK\$15,870.

Salaries and mandatory contributions are indexed to CPI, but real purchasing power of retirement income depends on whether investment returns exceed inflation. A 1% fee differential over 45 years can materially reduce the final balance (e.g., Peter's balance falls from HK\$4.99M at 1% fees to about HK\$4.32M at 1.5% fees in a sensitivity check).

Table 7: TVC required to reach target replacement ratio

Member	Current RR	Target RR	Extra Monthly TVC (HK\$)
Peter	36.0%	60%	1,582
Peter	36.0%	70%	2,242
Mary	9.1%	60%	13,205
Mary	9.1%	70%	15,799

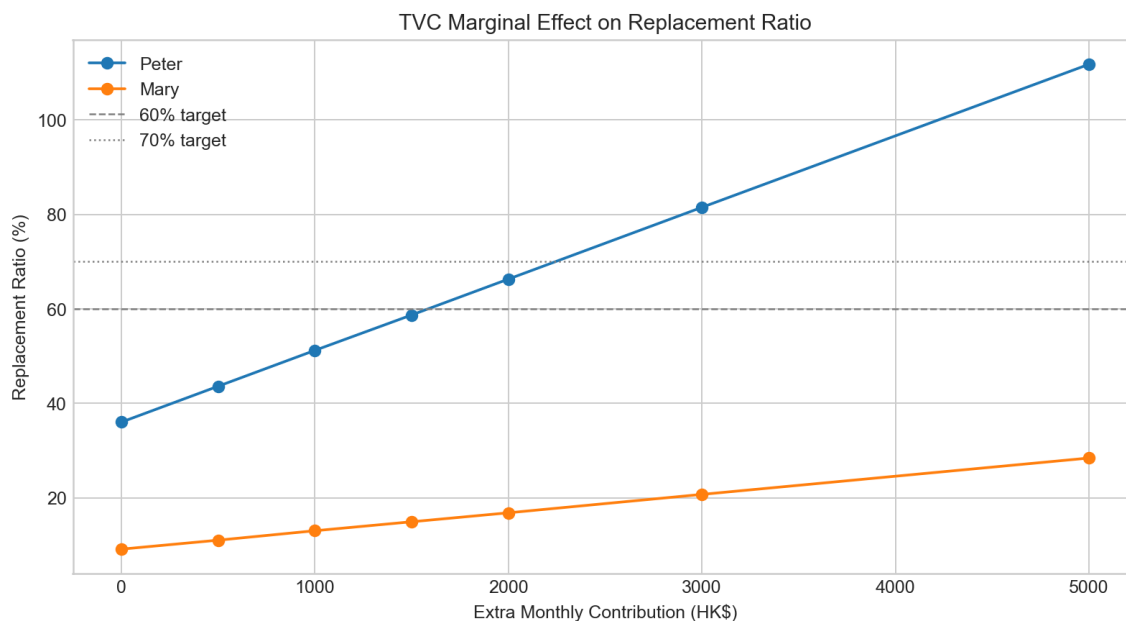


Figure 3: Marginal effect of TVC on replacement ratio

Mary urgently needs substantial TVC or other income sources. Peter still requires about HK\$1,600 per month extra to reach 60% replacement, achievable within the TVC tax-deductible limit (HK\$60,000/year $\approx$ HK\$5,000/month). By contrast, Mary's required TVC of HK\$13,205/month far exceeds the TVC cap; only the first HK\$5,000/month enjoys tax relief—the remainder must come from non-deductible VC or third-pillar savings.

(c) Supporting Retirement Under DC with the Social Security System

1. Hong Kong Three-Pillar Framework

Unlike Canada's CPP or US Social Security, Hong Kong has **no universal earnings-related public pension**. Government support is means-tested and **layered**:

- **First pillar** provides a floor for low-asset, low-income seniors (CSSA strictest; OALA asset/income-tested; OAA more universal but smaller)
- **Second pillar** (MPF) is the mandatory base for all employed persons
- **Third pillar** is the main lever for maintaining pre-retirement living standards

For middle-income workers such as Peter and Mary, first-pillar benefits are largely **unavailable** when MPF balances are counted in asset tests—retirement income must come from MPF plus voluntary savings, not from simple “stacking” of pillars.

2. Old Age Living Allowance (OALA)

Eligible Hong Kong residents aged 65+ may receive Normal OALA (HK\$2,065/month) or Higher OALA (HK\$4,185/month) in 2025/26, subject to income and asset tests. For a single person, the asset limit for both tiers is approximately HK\$401,000. **MPF accrued benefits not yet withdrawn are counted as assets.** This creates a **crowding-out effect**: the more DC wealth a member accumulates, the less likely they qualify for first-pillar support—the two are not simply additive.

3. Combined Retirement Income

Table 8 reports **actual** eligibility. Peter (MPF balance HK\$4.99M) and Mary (HK\$851K) both exceed the asset limit and therefore receive **no OALA** when MPF is counted. Table 9 shows hypothetical amounts *only if* assets were below the limit (illustrative contrast).

Table 8: Retirement income after OALA asset test (MPF counted)

Member	MPF Balance	Asset Limit	MPF Income	OALA	Combined RR
Peter	4,996,362	401,000	16,655	0	36.0%
Mary	850,862	401,000	2,836	0	9.1%

Table 9: Hypothetical OALA (if MPF not counted—not applicable to Case III)

Member	Scenario	MPF	OALA	Total	Combined RR
Peter	Normal OALA	16,655	2,065	18,720	40.5%
Peter	Higher OALA	16,655	4,185	20,840	45.0%
Mary	Normal OALA	2,836	2,065	4,901	15.7%
Mary	Higher OALA	2,836	4,185	7,021	22.5%

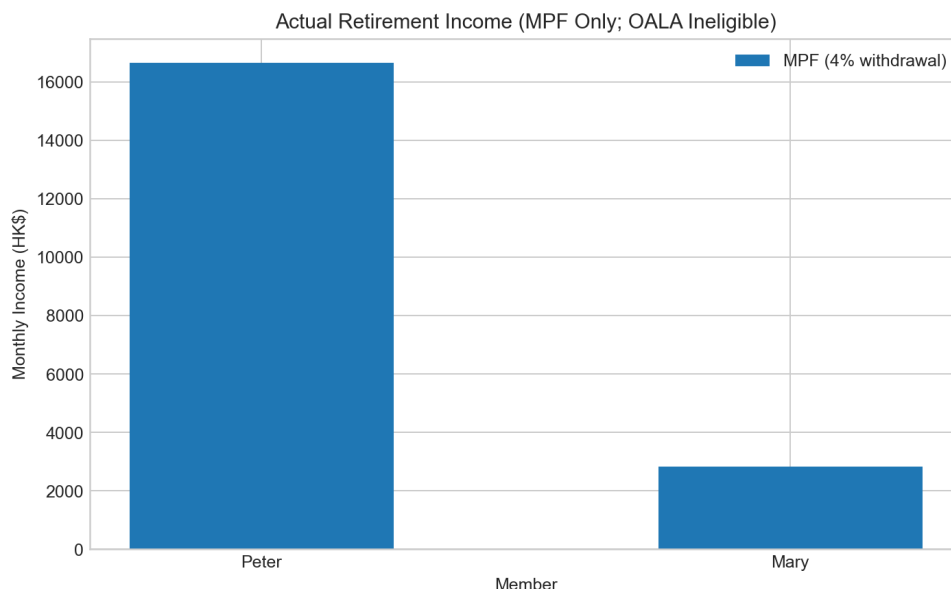


Figure 4: Actual retirement income (MPF only; OALA ineligible under asset test)

4. Illustrative Strategies

Peter (retirement 2045):

- MPF provides a solid base (36% replacement ratio) but OALA is **ineligible** given expected MPF assets.
- Add TVC of about HK\$1,600 per month to reach 60% replacement from MPF alone (within the TVC tax cap).
- Adopt a **lifecycle glide path**: high equity allocation when young, gradually shifting to bonds/capital-preservation funds after age 55 to mitigate sequence risk—rather than holding 100% global equity for 45 years.
- Annuitise part of the MPF balance to create a guaranteed lifetime income floor against longevity risk.

Mary (retirement 2025):

- Mandatory MPF alone is severely inadequate (9.1% replacement ratio); OALA is **ineligible** (balance HK\$851K > HK\$401K limit).
- Delaying retirement to age 70 raises the replacement ratio only modestly (to 10.9% in our simulation)—extra years of contribution help, but cannot offset 20 years of lost compounding.
- Required extra savings far exceed the TVC tax cap; combine non-deductible VC, third-pillar savings, and possible part-time work.

- Annuitisation and delayed withdrawal can still improve income security given limited account size.

A realistic plan for Mary—MPF (HK\$2,836) plus third-pillar savings or part-time income (e.g., HK\$5,000/month)—yields about HK\$7,836 per month, still below the 60% target of HK\$18,706. Even the hypothetical Higher OALA scenario (Table 9) reaches only 22.5%.

Table 10: Mary: delayed retirement sensitivity

Retirement Age	Contribution Years	MPF Balance (HK\$)	Replacement Ratio
65	25.1	850,862	9.1%
67	27.1	930,635	9.6%
70	30.1	1,128,770	10.9%

5. Conclusions on Extra Contributions

Table 11: Summary recommendations

Question	Peter	Mary
Is mandatory MPF sufficient?	No (36%)	No (9.1%)
Should they add TVC?	Yes, moderate	Yes, urgent
Can social security fill the gap?	No (asset test)	No (asset test)

Hong Kong’s DC system shifts retirement responsibility to the individual. Early, consistent contributions—as in Peter’s case—are the most powerful lever. Social security (OALA) provides a safety net for **low-asset** retirees but is crowded out by MPF accumulation for the middle class. **Both Peter and Mary should make extra contributions; the urgency and required amount are far greater for Mary.**

6. Core Conclusions

1. Mandatory MPF alone cannot maintain pre-retirement living standards; voluntary savings are essential.
2. **Contribution start date** is the dominant determinant of DC outcomes—time compounding outweighs fund selection.
3. Hong Kong’s first pillar provides a floor, not a supplement, for middle-income members with meaningful MPF balances.
4. A complete retirement plan combines mandatory MPF, voluntary contributions (TVC within tax limits), third-pillar savings, lifecycle de-risking, and partial annuitisation.

Key Formulae

Future value with CPI-linked contributions (monthly simulation):

$$FV = \sum_{t=1}^n PMT_0(1+g)^{t-1} \prod_{k=t+1}^n (1+r_k) \quad (1)$$

where PMT_0 is the initial monthly contribution, g is the annual contribution growth rate (CPI), r_k is the net return in period k , and n is the number of contribution periods.

When r and g are constant and $r \neq g$, the growing-annuity closed form is:

$$FV = PMT_0(1+r) \cdot \frac{(1+r)^n - (1+g)^n}{r-g} \quad (2)$$

Replacement ratio with a 4% withdrawal rate:

$$RR = \frac{FV \times 0.04}{Salary_{final}} \quad (3)$$

Required monthly TVC to reach target replacement ratio ρ^* : solve for PMT_{TVC} such that

$$\rho^* = \frac{0.04 \times FV(PMT_0, g, r, n) + 0.04 \times FV(PMT_{TVC}, g, r, n')}{Salary_{final}} \quad (4)$$

where n' is the number of periods over which extra contributions are made (here, the full contribution horizon). Table 7 values are obtained by binary search on this condition in the simulation model.

Monthly income gap to a 60% target:

$$Gap_{monthly} = 0.60 \times \frac{Salary_{final}}{12} - \frac{FV_{MPF} \times 0.04}{12} \quad (5)$$

References

- [1] Mandatory Provident Fund Schemes Authority, *MPF Fund Performance Platform*, <https://mfp.mpfa.org.hk/>.
- [2] Social Welfare Department, *Old Age Living Allowance*, <https://www.swd.gov.hk/oala/>.
- [3] Census and Statistics Department, *Composite Consumer Price Index*.
- [4] Investopedia, *Individual Retirement Account (IRA)*, <https://www.investopedia.com/terms/i/ira.asp>.

Appendix: Simulation Reproducibility

All numerical results are generated by the Python pipeline in `case3/`. To reproduce tables and figures:

```
cd case3
python3 main.py
python3 src/validate.py
```

Output files are stored in `case3/output/`. Assumptions are documented in `case3/data/assumptions.md`.